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Human vs. AI counseling: College students' perspectives

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A B S T R A C T

Transitioning to college life while navigating the complexities of emerging adulthood can be stressful. In some instances, it may even lead to the onset of mental health problems or the exacerbation of existing issues. While therapeutic resources are typically available in tertiary educational contexts, social stigma may lead to service underutilization. Additionally, high student-to-therapist ratios can create bottlenecks to access when such services are sought. Offering an adjunct to traditional campus counseling services, AI chatbots can potentially address such issues. Chatbots can provide flexible, accessible, anonymous, and cost-effective first-line support, improving access and extending traditional treatment methodologies. This study evaluates college students' perceptions ($N = 224$) of an AI chatbot (Pi) designed to emulate supportive and empathetic interactions characterized by active listening. Participants blindly assessed transcripts from active listening interactions between a client and Pi versus interactions between a client and a human counselor/therapist. The results indicate that participants could not distinguish between the human-human and human-AI counseling transcripts, answering correctly only 47.5% of the time. Moreover, participants gave higher quality ratings to the human-AI counseling transcripts than the human-human ones. These findings provide tentative support for the user-acceptability of relational AI chatbots during the counseling process's early phases (active listening and problem exploration).

1. Introduction

Attending college can be an exciting and challenging time for many students. College students may encounter various difficulties within both university and personal life, including challenges in relationships, health, future career planning (Pedrelli et al., 2015), and the need to form new social support systems (Albright and Hurd, 2018). Increasing evidence indicates that the mental health of college students is declining. Within this context, it is easy to understand why mental health problems are widespread among college students (De Girolamo et al., 2012; Ibrahim et al., 2013; Blanco et al., 2008). One in three students is prone to one or more mental health problems (Hunt & Eisenberg, 2010; Auerbach et al., 2018; Bruffaerts et al., 2018). During the 2020–2021 academic year, more than 60% of US students had at least one clinically significant mental health issue, according to validated screening tools (Lipson et al., 2015). These problems can lead to a long-term harmful impact on students' physical and mental health when they progressively build up without being resolved (Wang & Fu, 2023). Although many universities offer students counseling to provide emotional support (Cerolini et al., 2023), many students are reluctant to seek help due to the low perceived need, attitudinal barriers, and the lack of mental health education (Liu et al., 2022). Another barrier is that more than 80% of universities' low therapist–student ratio limits the opportunity to provide sufficient therapeutic sessions (Gagnon & Mattingly, 2016).

The increasing accessibility to technology has created new opportunities by providing digital mental therapy (Fitzpatrick et al., 2017). Recent progress in Artificial Intelligence (AI) technology has led to the development of chatbots capable of offering counseling and mentoring services (Gaffney & M.S, 2019). Chatbots are computer programs designed to facilitate communication with humans using natural language, eliminate time and space constraints, enabling users to interact with them anytime and anywhere (Bradeško & Mladenić, 2012). Chatbots can bridge treatment gaps, reduce client waiting times, and deliver lower-cost interventions (Cilliers et al., 2010). Chatbots used for mental health can act as "virtual friends" to help students vent their emotions by enabling them to communicate authentic emotions that are difficult to convey in person. Indeed, chatbots used for mental health are rapidly expanding in number, functions offered, and popularity and have been demonstrated to have a significant adherence rate (Vaidyam et al., 2019).

Many previous studies on mental chatbots mainly focused on constructing chatbots that adopt Cognitive Behavioral Therapy (CBT) principles (Koulouri et al., 2022). Little research demonstrates the

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effectiveness of mental chatbots. Among them is a qualitative study with 10 participants that compared chatbot-based therapy with human therapists (Bell et al., 2019). The study found chatbot-based therapy less useful and enjoyable than human therapists. Other studies have focused on students' acceptance of mental chatbots (Koulouri et al., 2022). However, the efficacy of chatbots in alleviating conditions such as depression, distress, and stress is still unclear, as there is no definitive proof of their clinical benefits (Abd-Alrazaq et al., 2020). Further, a few studies evaluated the perception of mental chatbots from the patients' perspective. Amongst this scant research are two studies that evaluated the satisfaction (Andrade-Arenas et al., 2024) and usability (Boyd et al., 2022) of mental chatbots. These studies contributed to the body of knowledge but used a low number of participants, which does not provide the needed statistical rigor.

This study aims to compare college students' perceptions of counseling provided by human counselors versus the counseling generated by Pi, a supportive and empathetic chatbot (Inflection, 2023). The aim of this study is motivated by the increasing integration of digital mental health interventions, particularly AI chatbots, within the mental health support ecosystem. As the demand for accessible and scalable mental health services grows, understanding how individuals perceive the effectiveness of AI-based counseling compared to traditional human counseling becomes essential. We also targeted college students as an important demographic, often experiencing heightened stress and mental health challenges. To address the aim of the study, we identified three primary objectives. First, the study seeks to determine whether college students can differentiate between counseling interactions produced by Pi and those generated by human therapists during the initial stage of psychotherapy (problem exploration). Second, the research aims to assess students' satisfaction with Pi-generated counseling compared to human-generated counseling. Third, the study compares students' perceptions of Pi-generated versus human-generated counseling. This comparison provides valuable insights into the effectiveness of AI in replicating complex human interactions within a therapeutic context. Furthermore, identifying differences between Pi-produced and human-human counseling interactions can inform developers in enhancing AI's emotional intelligence and conversational capabilities.

Our study involved 224 college students, who were asked to distinguish between counseling transcripts created by a human client interacting with Pi (human-AI) and those generated by traditional interactions between human counselors and human clients (human-human). Additionally, the students were asked to rate their satisfaction with the transcripts. The results of this study have significant implications for the evolving role of AI as a supplementary tool to traditional psychotherapeutic practices and service delivery.

This study presents a significant strength by offering an empirical evaluation—quantitative and qualitative—of content produced by AI systems within the critical realm of mental health. Given the expanding integration of AI across diverse aspects of everyday life, assessing the quality and impact of these systems is crucial. This research builds on our recent work detailed in Kuhail et al. (2024), which analyzed professional therapists' perspectives on AI-driven counseling. However, the current study expands its scope to explore the perspectives of the general population, an essential aspect for understanding the broader societal implications and acceptance of AI in mental health services.

The subsequent sections of this paper are structured as follows: Section 2 presents the related work, Section 3 provides a detailed description of our methodology, Section 4 presents our findings, Section 5 offers a discussion of these findings, and finally, Section 6 concludes our paper.

2. Related work

Student mental health among university students is a major global concern due to its pervasive impact on academic performance and overall students' lives (Lipson et al., 2019). According to Pham et al. (2019), university students have a higher frequency of mental health issues than the overall population. Moreover, the Healthy Minds Study (HMS) (2020) reported that more than 60% of surveyed students met the criteria for one or more mental health issues. Students reported mental health issues for a variety of reasons, including navigating the transition into adulthood, adjusting to university life, living away from family and friends, financial hardship, and lack of self-determination (Chang & Chang, 2011; McLafferty et al., 2017; Pham et al., 2019). As stated by Malik et al. (2023), anxiety ranked as the top concern among university students (41.6%), depression (36.4%), and suicidal thoughts (16.1%). Additionally, these rates have increased over the past few decades; for example, Chinese university students surveyed on the Self-Rating Depression scale grew by about 6 points (Feng et al., 2020). Therefore, addressing mental health issues requires extensive mental health services. Universities can play a critical role in mitigating the impact of mental health issues on their students. Using university mental health services could address students' issues and help them thrive academically, emotionally, and socially. According to a study by Vidourek et al. (2014) 682 university students perceived many benefits of using mental health services. These benefits include enhanced mental health, decreased stress, resolving students' problems, self-awareness and personal growth, happiness, and life satisfaction. As a result, the importance of mental health for university students cannot be overstated, as it significantly influences the overall quality of their lives during their educational journey.

Many universities have free and confidential counseling services and mental health advisers professionally qualified to provide prompt access to therapist care. However, many studies have indicated a low prevalence of mental health service utilization among university students (Cadigan et al., 2019; McLafferty et al., 2017; Eisenberg et al., 2007; Tien et al., 2021). Increasing the number of counseling staff in universities is one of the solutions to adequately addressing the growing demand for mental health support among students to reduce waiting times and enable more personalized care tailored to individual student needs. However, increasing the number of therapists in universities can be challenging due to various factors such as budget constraints and finding expert therapists and appropriate support systems (e.g., office space, technology, and administrative staff). To meet the students' mental health needs, additional forms of mental health services are needed (Bennion et al., 2020). As such, digital mental health offers a promising solution to address student mental health challenges through convenient access to resources and services that can assist in managing their mental well-being anytime, anywhere. Digital mental health can address various aspects, such as assessing the situation of students, supporting students informationally and emotionally, preventing some mental health issues, and offering treatment (Wies et al., 2021).

Chatbots have shown promise in mental health as they have been widely used to provide people with mental health concerns with a variety of social support services, including informational, emotional, instrumental, and evaluation help (Kim et al., 2018). A chatbot is a computer program designed to interact and work with people in written, spoken, and visual languages (Adamopoulou & Moussiades, 2020). Chatbots have proved useful in improving mental health issues for people of different backgrounds and demographics. For example, chatbots have supported students experiencing anxiety and stress (Fitzpatrick et al., 2017) and workers in the health care system who need emotional support (Judson et al., 2020). Moreover, chatbots have been used for problem-solving among nonclinical older adults (Bennion et al., 2020) and people living in rural areas (Potts et al., 2021). Findings of previous research indicate that people are increasingly accepting chatbots as a means of aiding various mental health issues, showing enhanced outcomes in mental health domains. Presently, there are a lot of chatbot apps on app stores with more than 500,000 downloads (Haque & Rubya, 2023).

For mental chatbots to be considered efficacious and beneficial, robust and substantial evidence derived from multiple empirical studies

must support the effectiveness and satisfaction with their therapeutic interventions. Extant literature shows that a few chatbots have proven efficacious. For example, Fitzpatrick et al. (2017) assessed the efficacy of the Woebot chatbot in giving CBT to university students with depression and anxiety. and found that the chatbot notably reduced depressive symptoms. Gabrielli et al. (2021) evaluated Atena, a psychoeducational chatbot supporting healthy coping with stress and anxiety, and found a significant decrease in anxiety and stress among students. Beatty et al. (2022) examined Wysa, a chatbot that leverages evidence-based cognitive-behavioral techniques. The authors found that the students' therapeutic alliance enhanced over time, and Wysa can be comparable to human-delivered face-to-face psychotherapy. Klos et al. (2021) assessed students' perceptions of Tess, a chatbot that supplies brief text chats as comprehensive support (e.g., psychoeducational content, and emotional support responses). The author found that students reported a significant enhancement in anxiety symptoms. Consequently, the literature highlights the effectiveness of a few chatbots in supporting various mental health issues among university students.

Although chatbots are being integrated into mental health services and social support, rigorous assessment remains a critical gap in our understanding of the efficacy and usability of these innovative tools. A recent review highlights that the evidence for chatbots' effectiveness in alleviating conditions such as depression, distress, and stress is still limited (Abd-Alrazaq et al., 2020). Further, the evaluation reported in the literature is primarily conducted by the developers of the chatbots (e.g., (Cameron et al., 2019; Daley et al., 2020; Fitzpatrick et al., 2017). Evaluations of chatbots by their developers may be subject to inherent biases. Consequently, there is a pressing need to establish standardized evaluation metrics. Independent third-party evaluation could offer impartial and insightful analyses of these chatbots, enhancing their credibility and utility. Further, it is essential to evaluate therapy from the patients' perspective to understand therapeutic alliance and possibly its impact on attrition (Murphy et al., 2022). Only a few studies have assessed the patients' satisfaction with mental chatbots, and while the results are encouraging, these studies tend to recruit a low number of participants (Andrade-Arenas et al., 2024; Boyd et al., 2022) and, as such, lack statistical rigor. Indeed, ongoing research must examine and compile evidence on their acceptability thoroughly (Abd-Alrazaq et al., 2020). Among the few studies examining mental health chatbots, our recent work (Kuhail et al., 2024) focused on how professional therapists (N = 63) viewed an empathetic chatbot. This study, however, shifts the focus from professionals to the public. It assesses how everyday users (in this case college students) perceive and are satisfied with empathetic chatbots, providing a broader view of how these AI tools are accepted across different groups.

This study seeks to evaluate the efficacy of chatbots used in mental health counseling by assessing the ability of human users to differentiate between human and AI-based counseling. Additionally, it compares user perceptions and satisfaction with both types of counseling. Distinguishing between AI-generated and human-generated text is essential for maintaining integrity and preventing misuse across various fields, including journalism and education (Alamleh et al., 2023). Recent research articles have highlighted differences between AI and human-authored text characteristics; for instance, human-authored text typically displays higher perplexity, increased grammatical errors, and more varied n-gram distributions compared to AI-generated texts (André et al., 2023). Conversely, AI-generated texts are often more analytical and descriptive but score lower on readability than human texts (Markowitz et al., 2024). Subsequently, several AI tools have been developed to detect AI-generated content (Hayawi et al., 2024). Furthermore, emerging literature explores how humans perceive AI-generated text. In journalism, a recent study found no significant differences in the perceived credibility and trustworthiness between AI-written and human-written texts in simple and concise formats (Lermann Henestrosa et al., 2023). To conclude, although there is a growing body of research comparing AI and human text in areas such as education and journalism, there remains a gap in understanding human users' perceptions and ability to discern AI from human counseling in the mental health sector. Investigating whether users can distinguish AI from human text in counseling contexts will enhance our understanding of how AI-generated counseling compares to human interaction and its potential as a supplementary tool to human therapists.

3. Methodology

3.1. Procedure and research design

3.1.1. Study procedure

The study employed a structured survey process, beginning with obtaining informed consent from student participants recruited across multiple universities in Saudi Arabia. Participants received email invitations describing the study's purpose, which involved evaluating therapy transcripts to assess therapeutic effectiveness and the therapist's perceived human likeness. They were informed of their voluntary participation and right to withdraw at any time.

After consenting, participants were directed to an online survey platform, which collected quantitative and qualitative data. They were presented with a counseling scenario, an anonymized therapy transcript randomly assigned from a human-human or human-AI transcript. Participants read and evaluated the transcript, providing feedback on the counseling's effectiveness and satisfaction with the interaction. They identified whether they perceived the interaction as human-AI or human-human and provided reasons for their perception. This process was repeated with a second scenario using the same evaluation criteria. Afterward, participants submitted their demographic information, and all responses were collected anonymously to ensure privacy and confidentiality. On average, the survey took 20–30 min to complete. The questionnaire combined Likert-scale and open-ended questions to gather comprehensive feedback on the counseling transcripts.

3.1.2. Quantitative research design

This study employed a between-subjects experimental design to assess participants' perceptions of AI-generated vs. human-developed counseling scenarios. These scenarios were based on two types of interactions: one between human clients and human counselors (human-human) and the other between human clients and an AI chatbot named Pi (human-AI). In this study, we chose Pi because it is designed to provide emotional support to users through a chatbot capable of maintaining interactive text or voice-based dialogues (Hu, 2023). This chatbot simulates active listening and aims to deliver an engaging user experience characterized by expressions of kindness, a diplomatic approach to sensitive topics, and the inclusion of humor, thereby fostering a supportive and engaging interaction environment.

We selected four human-human counseling transcripts from three Cognitive Behavioral Therapy (CBT) books to maintain consistency. These transcripts focused specifically on active listening and did not include any specific CBT techniques. Each transcript began with an initial statement from the human client. For each of the four human-human counseling transcripts, we derived human-AI counseling as follows. We inputted the same client's opening statement from each human-human transcript into Pi, allowing Pi to respond and continue the conversation. However, the interactions with Pi naturally diverged from those in the original human-human transcripts because Pi's responses were different from the human-human transcripts. Fig. 1 provides an example of one of the original CBT scenarios from the books and the corresponding scenario that evolved in conversation with Pi. For further reference, we have included the details of the scenarios together with the references to the CBT books they were extracted from in the appendix.

Participants were randomly allocated into four groups, each assigned to evaluate two specific scenarios. Efforts were made to maintain relatively equal group sizes during the assignment process. However, minor

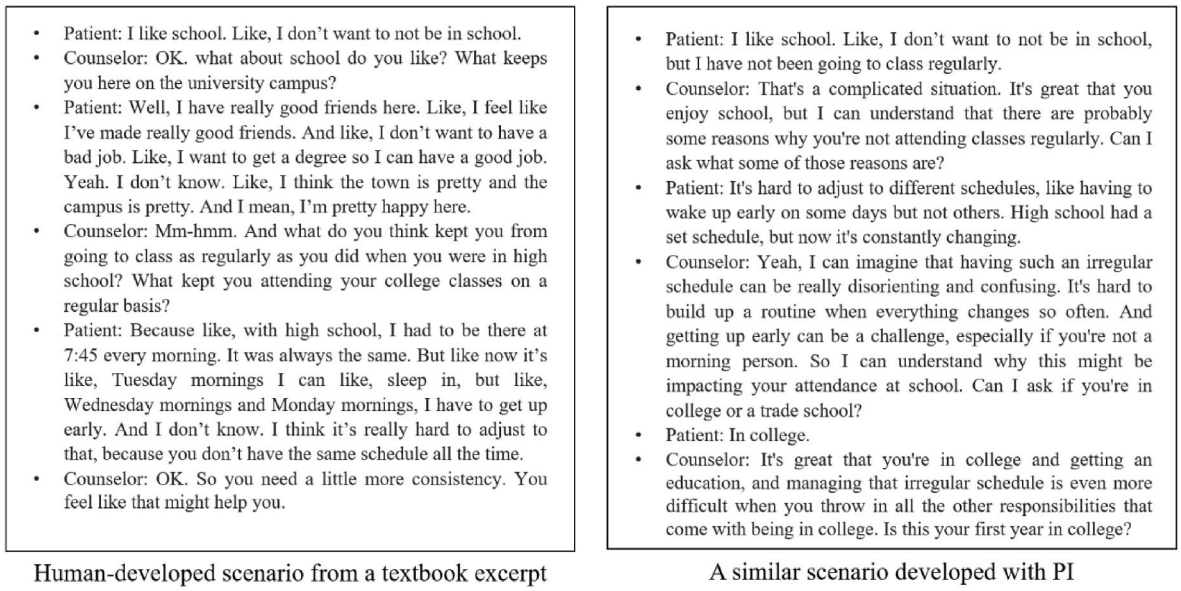


Fig. 1. A comparison between textbook-developed counseling and AI-assisted counseling.

variations in group sizes occurred due to participant attrition throughout the study. In total, our study included eight scenarios: four involving human-human counseling and four involving human-AI counseling. To prevent direct comparisons between human-human and the corresponding human-AI scenarios and to minimize bias, the assignment of scenarios to each group was carefully controlled. We included all possible permutations of scenario combinations to ensure comprehensive coverage and balance. Each group received a different combination of human-human and human-AI scenarios, as detailed in Table 1. Specifically, Group 1 evaluated two human-human scenarios (scenarios H1 and H2), Group 2 reviewed a human-human scenario followed by a human-AI scenario (scenarios H3 and A1), Group 3 started with a human-AI scenario before assessing a human-human scenario (scenarios A2 and H4), and Group 4 analyzed two AI-generated scenarios (A3 and A4).

3.1.3. Qualitative research design

Alongside quantitative measures, the study included a qualitative component using a thematic analysis approach to gain deeper insights into participants' perceptions of counseling scenarios. Participants were asked open-ended questions to explain their reasons for identifying a counseling scenario as human-human or human-AI. These responses were collected to understand their judgments' nuances and explore factors influencing perceived human likeness and therapeutic satisfaction. The qualitative data were gathered from participants' detailed feedback on the counseling interactions. This open-ended feedback was analyzed thematically to identify recurring patterns, themes, and insights related to the chatbot's interaction style and the perceived differences between human and AI counselors.

3.2. Sample

Table 2 shows the demographic data of the study population. The sample is 224 university students, with a majority of female participants, with males constituting a smaller fraction of the total. We have used convenience sampling (Sedgwick, 2013) due to the accessibility of students within the institutions of the authors. Our sample size is limited by the number of students who agreed to participate, but the size is comparable to the sample sizes reported in studies of similar fields (Novielli et al., 2023; Peterson & Merunka, 2014). The age distribution is predominantly within the 18–25 year range, with minimal representation from the 26–35 and 46–55 year brackets. Regarding chatbot usage, responses vary from 'always' to 'never,' indicating diverse engagement levels with technology. The educational qualifications of participants are largely at the high school level, followed by those holding bachelor's degrees, with very few possessing master's or PhD degrees. The majors participants pursue are primarily in computing, natural sciences, engineering, economics, and other fields.

Table 1
Setup of groups and configuration of scenarios.

Group No.	Count	First Scenario	Second Scenario
1.	61	H1	H2
2.	47	H3	A1
3.	47	A2	H4
4.	69	A3	A4

Table 2
Demographic characteristics of the student's sample.

Characteristic	Count
Gender	Male (N = 69, 30.8%), female (N = 155, 69.2%)
Age	18-25 (N = 222, 99.1%), 26–35 (N = 1, 0.4%), 36–45 (N = 0, 0%), 46–55 (N = 1, 0.4%)
Chatbot usage	Always (N = 29, 12.9%), frequently (N = 46, 20.5%), sometimes (N = 89, 39.7%), rarely (N = 33, 14.7%), never (N = 27, 12%)
Highest qualification	High school (N = 145, 64.7%), bachelor (N = 74, 33%), master (N = 3, 1.3%), PhD (N = 2, 0.8%).
Major	Computing (N = 152, 67.8%), natural sciences (N = 9, 4%), engineering (N = 15, 6.7%), economics (N = 15, 6.7%), medicine (N = 6, 2.6%), sociology (N = 4, 1.7%), arts (N = 3, 1.3%), music and film industry (N = 2, 0.8%), public relations (N = 1, 0.4%), undecided (N = 17, 7.5%).

3.3. Measurements and reliability

The independent variable in this study is the type of counseling provided, which is operationalized as either a counseling interaction between a human therapist and a human client or a counseling interaction between an AI system and a human client. The dependent variables are (1) participants' perceptions of the counseling process and (2) participants' overall satisfaction with the counseling experience.

We used questionnaire questions to measure the dependent variables

and collect qualitative data. The questionnaire can be found in the appendix (Fig. 2). We measured participants' perceptions of the counseling using six statements supported by related studies (Question 1, Fig. 2). The statements judged these elements: trustworthiness, warmth and empathy, organization and competence, and interest in the conversation. These elements are supported by various studies, such as (Ackerman & Hilsenroth, 2003; Seewald & Rief, 2023). The participants were asked to rate to what extent they agreed with the statements on a 1–5 scale (1 = strongly disagree, 5 = strongly agree). The statement "The counselor seemed trustworthy." measured trustworthiness. The statement "The counselor seemed warm" and "The counselor seemed empathetic." measured warmth and empathy. The statements "The counselor seemed organized." and "The counselor seemed competent." measured organization and competence. Finally, the statement "The counselor seemed interested in the conversation." measured interest in the conversation.

We also measured participants' satisfaction with the counseling by asking, "Based on this short extract, If that was you in the session, how satisfied would you have been?" (Question 2, Fig. 2). The participants rated their satisfaction on a 1–5 scale, where 1 means highly dissatisfied and 5 means highly satisfied.

We also asked the participants to determine the scenarios they read as "The product of human-human interaction (Human)" or "The product of human-artificial intelligence interaction (AI)." (Question 3, Fig. 2). To obtain qualitative data, we asked the same participants in the questionnaire to explain why they chose to collect qualitative data (Question 4, Fig. 2).

We calculated Cronbach's alpha to measure the internal consistency of the items used to evaluate counseling evaluation. The Cronbach's alpha is 0.91, indicating high internal consistency. Skewness and Kurtosis values were calculated for each data column to test the normal distribution of the data. The skewness and kurtosis values for all the data columns were observed between −2 and +2 (Byrne, 2016; George & Mallery, 2016; Hair Jr et al., 2009), indicating that the data is reasonably close to being normally distributed.

4. Results

4.1. Descriptive statistics

The descriptive statistics for participant responses regarding counseling scenarios show a preference for human-AI counseling over human-human counseling (Table 3). Python's pandas library was utilized to compute the descriptive statistics. The average score for participants' perceptions of human-AI counseling was higher (mean = 3.64) with a standard deviation (std) of 0.92, indicating a generally positive view with moderate response variability. In comparison, human-human counseling had a lower average score (mean = 3.12) with a similar level of response variability (std = 0.89). Satisfaction scores followed a similar pattern, with human-AI counseling scoring higher on average (mean = 3.46, std = 1.22) than human-human counseling (mean = 3.09, std = 1.15), though with greater variability in satisfaction levels.

While determining if the human-human counseling was human, they made 98 correct and 118 incorrect guesses. Similarly, participants made 115 correct and 117 incorrect guesses for AI-human counseling, indicating a nearly even split in their ability to distinguish between AI-generated and human-developed counseling.

Table 3
Descriptive statistics.

Item	Results
Perception of human-AI counseling	mean = 3.64, std = 0.92
Perception of human-human counseling	mean = 3.12, std = 0.89
Satisfaction with human-AI counseling	mean = 3.46, std = 1.22
Satisfaction with human-human counseling	mean = 3.09, std = 1.15
Determining if the human-human counseling was indeed human-human	Correct guess (N = 98), Incorrect guess (N = 118)
Determining if the human-AI counseling was indeed human-AI	Correct guess (N = 115), Incorrect guess (N = 117)

4.2. Statistical tests

We used Python's stats library to run the statistical tests. We conducted a chi-squared test of independence, a test conducted to determine whether there is a statistically significant association between two categorical variables. In our study, the test examined the relationship between participants' ability to detect human-AI and human-human counseling. The chi-squared test statistic was calculated to be χ^2 (1) = 0.9539. The p-value associated with this chi-squared statistic was 0.3287, indicating that it is greater than the chosen significance level 0.05. The data do not support a significant association between participants' ability to detect human-AI counseling and their ability to detect human-human counseling. In practical terms, this suggests that participants who can detect one type of counseling scenario do not appear to have a significant advantage in detecting the other type.

We conducted a Mann-Whitney *U* test to compare students' perceptions of human-AI counseling and human-human counseling to test Hypothesis H1, which posits that there is a difference in students' perceptions of these two counseling types (Table 4). The Mann-Whitney *U* test, a non-parametric statistical method, is appropriate for comparing differences between two independent groups when the data does not follow a normal distribution. This was confirmed by a Shapiro-Wilk test, which indicated non-normality in the data (W = 0.977, p-value = 2.26). Levene's test assessed the homogeneity of variances between two groups: Group 1, which evaluated students' perceptions of human-AI counseling, and Group 2, which assessed perceptions of human-human counseling. The test statistic was 0.637, with a p-value of 0.425. These results indicate that there is no significant difference in the variances between the two groups, thereby supporting the assumption of homoscedasticity.

The Mann-Whitney *U* test revealed a significant difference in participants' perceptions of human-AI and human-human counseling (U-value = 16764, p-value = 1.34×10^{-9}*), with participants rating AI-generated counseling scenarios more favorably on average than those based on human-human interactions.

We also conducted a Mann-Whitney *U* test to compare students' satisfaction with human-AI counseling and human-human counseling to test Hypothesis H2, which posits that there is a difference in students' satisfaction between these two types of counseling (Table 4). Here, too, the Mann-Whitney *U* test was suitable due to the data's non-normality, as confirmed by a Shapiro-Wilk test (W = 0.905, p-value = 4.21). Levene's test showed no significant difference in variances between the two groups—Group 1, evaluating students' perceptions of human-AI counseling, and Group 2, assessing human-human counseling (test statistic = 0.011, p-value = 0.913). This supports the assumption of homoscedasticity.

The Mann-Whitney *U* test revealed a significant difference between participants' satisfaction with human-AI and human-human counseling (U-value = 20393, p-value = 0.0004). As previously mentioned, participants were more satisfied with human-AI than human-human counseling on average.

Table 4
Hypothesis testing results.

Hypothesis	U-value	p-value	Result
H1: There is a difference in students' perceptions of human-AI counseling compared to human-human counseling.	16764	1.34×10^{-9}	Supported
H2: There is a difference in students' satisfaction with human-AI counseling compared to human-human counseling.	20393	0.0004	Supported

4.3. Qualitative results

This section presents the themes for why participants determined whether a scenario was an interaction between AI and humans or humans and humans. We divided the themes into four parts: (1) The first part presents themes for students to correctly determine that the scenarios were indeed the interaction between humans and humans. (2) The second part presents themes for students incorrectly labeling human-human counseling as human-AI counseling. (3) The second part presents themes for students correctly determining that the counseling was indeed the interaction between humans and AI, and (4) The fourth part presents themes for students incorrectly labeling the human-AI counseling as human-human counseling. To identify the themes, we performed a thematic analysis following the approach outlined in (Braun & Clarke, 2022), which involved the following steps: (1) familiarizing ourselves with the data, (2) creating initial codes, (3) identifying potential themes, (4) defining the themes, (5) iteratively refining and reviewing the themes, and (6) compiling the results in the final report. No software was used for the thematic analysis.

Participants who correctly identified human-human counseling cited three themes: Human-like interaction, structured responses, and emotion and warmth in language (Table 4). In 63 instances, participants described qualities such as compassion and tailored responses, which led to a perception of human interaction. Within this theme, we identified subcategories like personalized empathy and depth of connection. For example, participants noted how therapists adjusted their communication based on the client's emotional state, demonstrating what could be described as adaptable emotional intelligence.

Structured responses, observed in 34 cases, involved the counselor's approach of seeking more in-depth reasons rather than hastily concluding. Subcategories within this theme included reflective listening and situational adaptability, where participants saw the counselor's responses adapting to the conversation's specifics. Lastly, Emotion and warmth in language, mentioned in 32 instances, were recognized by the presence of emotional depth in the communication, contributing to the perception of the interaction being more human-like. Subcategories in this theme include the counselor's ability to use and understand human emotions.

When participants mistakenly identified human-human counseling as human-AI counseling, four key themes emerged: AI-like and robotic responses, formal and disorganized interaction, lack of feeling and empathy, and limited human interaction (Table 5). The most frequently cited theme was AI-like and robotic responses, observed in 51 instances. Subcategories within this theme include conversational rigidity, repetitiveness, fakeness, lack of nuanced responses, premature problem-solving, and excessive punctuation. Disorganized interaction was noted in 37 cases. Subcategories within this theme included incoherence and topic jumping. The lack of feeling and empathy, cited in 36 instances, highlighted a perceived absence of sympathy and a simplistic question-and-answer format. Subcategories within this theme include the absence of warmth, emotional intelligence, and lack of empathy. Finally, limited human Interaction, observed in 28 cases, involved counselor responses that seemed impersonal and disconnected from truly understanding the individual. Participants identified subcategories such as a lack of personal connection and superficial advice.

Participants who correctly identified human-AI counseling highlighted four major themes: AI-like and robotic responses, lack of feeling and empathy, formal interaction, and limited engagement and interaction (Table 6). "AI-like and robotic responses" is the most prominent theme, with 73 instances. Subcategories included a lack of improvisation and pre-programmed-sounding responses, where the AI seemed to mirror the participant's input without offering new insights. Lack of feeling and empathy, noted in 62 cases, was identified by the therapist's absence of emotional expression. Subcategories included emotional flatness and insensitivity to participant emotions, contributing to the AI's perceived inability to connect on a human level. Formal interaction, mentioned 57 times, described the interactions as formal and concise. The subcategories within this theme included a formal, structured approach and brief responses that did not allow for deep conversational flow. Lastly, limited engagement and interaction were observed in 48 scenarios. Subcategories included a lack of emotional reciprocity and the absence of back-and-forth communication.

When participants incorrectly labeled AI-human counseling as human-human interactions, they did so based on themes of emotional involvement, engagement in dialogue, solution-oriented analysis, and rational support (Table 7). The most common theme, emotionally involved and empathetic, with 71 instances, described the AI counselor as warm and empathetic, mirroring human traits. Subcategories include the ability of the counselor to understand and use emotions, as well as the ability to empathize with clients. Engaged in detailed and personal dialogue, noted in 55 cases. Subcategories include the counselor's ability to provide reflections and the active involvement in the conversation. Suggesting solutions and deep analysis, cited 45 times. Subcategories include perceived human-like intelligence and analytical capability. Lastly, rational and supportive responses, identified in 34 instances. Subcategories include that the counselor seemed knowledgeable and supportive (Table 8 and Table 9).

5. Discussion, implications, and future research directions

The early phase of counseling involves problem exploration through active listening. This study contributes significantly to our understanding of AI's capabilities in replicating human therapeutic practices. The AI chatbot in the present study was able to perform this function, as well as the human therapist, in the estimation of prospective clients (healthy students). Blinded to the actual source, student raters could not differentiate between the transcripts of human therapists and AI and even expressed a significantly more favorable rating of the AI chatbot's performance. A major strength of this study lies in its empirical demonstration that AI can be perceived as equivalent to human therapists, prompting further inquiry into the underlying factors contributing to this perception. However, this finding begs the question of the factors contributing to their perception of the AI chatbot as human. One potential explanation is that advancements in AI have enabled chatbots to generate increasingly human-like text (Chen et al., 2024). These technological improvements allow AI systems to replicate the nuances of

Table 5
Themes for students correctly identifying the human-human counseling scenarios.

Theme	Count	Example
Human-Like Interaction	63	"The therapist seems more compassionate and more experienced"
Structured Responses	34	"In this case the counselor asked for more reasons rather than jumping into conclusions which often occurs when dealing with an AI"
Emotion and Warmth in Language	32	"It's felt like a human because of the level of emotions"

Table 6
Themes for students mistakenly identifying the human-human counseling scenarios as AI-human interactions.

Theme	Count	Example
AI-Like and robotic responses	51	"Short answers alongside sentences that have heavy use of punctuation."
Disorganized interaction	37	"The response is disorganized and never gives details"
Lack of feeling and empathy	36	"Because there is no sympathy method, just question and answer." "The therapist was very formal and lacked emotion. "
Limited human Interaction	28	"The counselor's responses don't sound like they really know the individual."

Table 7
Themes for students correctly identifying the human-AI counseling.

Theme	Count	Example
AI-Like and robotic responses	73	"Because the therapist was repeating what the student were saying and she didn't give advice or solutions to the problem."
Lack of feeling and empathy	62	"Because the therapist did not show emotions."
Formal Interaction	57	"Formal and brief"
Limited engagement and interaction	48	"The conversation is very one-sided; the counselor should be trying to get the patient's opinion."

Table 8
Themes for students mistakenly identifying the human-AI counseling scenarios as human-human ones.

Theme	Count	Example
Emotionally involved and empathetic	71	"The counselor was warm and had an empathetic touch to the conversation."
Engaged in detailed and personal dialogue	55	"The therapist talks to the patient as if he understands the issue and is trying to solve the problem."
Suggesting solutions and deep analysis	45	"A human being because he is deeply involved in the dialogue with the patient."
Rational and supportive responses	34	"He knows what he is talking about."

natural language, making it more difficult for users to distinguish between AI-generated responses and those from human counselors. However, a future research direction is understanding the factors that led to the findings. Was it the AI chatbot's language, empathy, or conversational style? Understanding these factors could inform the future development of more sophisticated and human-like chatbots.

Our findings are aligned with Aktan et al. (2022) and Kuhail et al. (2024), who reported a similar preference for an AI-based psychotherapist. However, it is important to note that this preference was based on a brief transcript focusing on one small aspect of the counseling process (problem exploration). However, the fact that AI performed so well suggests that this is indeed a fruitful area for further research and development.

Qualitative findings also emphasize the importance of considering differences in users' perceptions when designing chatbot counseling services. Adapting interventions leads to considering preferences that may optimize user experience and effectiveness. As a result, understanding these findings is crucial for designing effective counseling interventions and AI systems that meet users' expectations. This is in line with previous studies, for example the study by Laranjo et al. (2018).

A key strength of this study is the finding that participants struggled to distinguish between human-AI and human-human counseling, indicating the advanced capability of AI to mimic human interactions effectively. This suggests significant potential for wider acceptance and accessibility of AI-based support services. However, it also highlights the need for ethical transparency and trust in AI deployment in sensitive areas like mental health, something that other researchers are calling for (Giubilini, 2021).

Despite the promising findings, this study has several limitations that must be acknowledged for a balanced interpretation of its contributions. First, the study's reliance on written transcripts may not fully capture the complexity of real-life interactions, which include critical elements like body language and vocal cues—essential in face-to-face counseling but less significant in text-based online settings (Walther, 1996). Additionally, the brief, focused transcripts used in this study may only partially represent the broader and more dynamic context of a full counseling session, which often involves a variety of therapeutic techniques beyond active listening. Moreover, the sample consisted of Saudi college students, which may limit the generalizability of the results to other populations. Another obvious limitation of this study is that those judging the quality of the interaction and attempting to differentiate between AI and human therapists are laypersons with little or no knowledge of the counseling practice.

Although this is an interesting indication of AI's potential in counseling, further research is needed to determine whether similar outcomes would be observed in different cultural or educational contexts. These limitations suggest the need for cautious interpretation and underscore the importance of additional studies exploring the effectiveness of AI in more complex and extended therapeutic scenarios.

Despite the seemingly acceptable performance of AI in the present study, it is crucial to differentiate between the actual efficacy of AI chatbots for counseling and the students' perceptions and satisfaction with the chatbots. While a positive perception of AI chatbots does not necessarily equate to their actual efficacy, the findings are still promising. Patients' positive perceptions and satisfaction with AI chatbots for counseling can reduce attrition rates and encourage adoption (Nahum-Shani et al., 2022). In assessing actual effectiveness, future studies would need to consider outcomes at treatment-end and longer-term follow-up. However, even if AI proves less effective than human therapists, it could still perform an adjunctive or partnering role, supporting therapists with problem exploration and other more routine aspects of the process. Furthermore, AI may prove invaluable in providing out-of-hours support and improving access to early intervention.

Future studies on AI in counseling could extend the scope and complexity of the interactions studied to mimic full counseling sessions better, incorporating various therapeutic techniques beyond basic active listening. This would provide a more comprehensive assessment of AI's capabilities in real counseling contexts. Additionally, addressing the limitations of text-based interactions, future research could focus on multimodal communication, integrating AI's ability to process and simulate non-verbal cues such as facial expressions and vocal tones, which are crucial in face-to-face counseling (Norcross & Lambert, 2011). Ethical and privacy concerns are also critical; future research must rigorously address data security and informed consent, ensuring AI's use in counseling adheres to ethical standards (Morley et al., 2020). Moreover, evaluating AI's integration into healthcare systems and its real-world effectiveness can provide insights into its practical viability and improvement areas, helping bridge the gap between AI capabilities and clinical needs.

6. Conclusion

This study contributes to the body of knowledge on the potential of chatbots in mental health care, specifically exploring their role as a viable supplement to traditional counseling services. In this study, we involved 224 college students who blindly evaluated counseling session transcripts involving both an empathetic chatbot (Pi) and human counselors. Our results show that these students could not reliably distinguish the AI-generated transcripts from those created by human counselors, with a correct identification rate of just 47.5%, equivalent to random chance. This demonstrates a key strength of the AI chatbots' ability to closely approximate human responses in a therapeutic setting. Furthermore, students rated the human-AI counseling transcripts more positively than those from human-human sessions. These findings suggest that AI chatbots like Pi could serve as effective preliminary support for people with mild to moderate psychological issues, indicating a valuable direction for further research and application in mental health care.

However, while the study's strengths lie in demonstrating AI's capacity to blend into therapeutic environments, limitations also emerged. Despite AI chatbots' promising role in complementing traditional counseling, enhancing service accessibility, and reducing counselors' workloads, the need for improvements in AI chatbots' ability to mimic the nuanced emotional and conversational aspects of human

interactions is clear. Future studies should also explore multimodal communication to better assess AI's effectiveness in counseling settings. As AI technology advances, it is crucial to thoroughly evaluate its strengths and limitations in the context of mental health counseling.

CRediT authorship contribution statement

Mohammad Amin Kuhail: Writing – original draft, Supervision, Project administration, Methodology, Funding acquisition. **Nazik Alturki:** Writing – original draft, Methodology, Funding acquisition, Formal analysis. **Justin Thomas:** Writing – review & editing, Validation, Formal analysis. **Amal K. Alkhalifa:** Writing – original draft, Methodology, Investigation, Formal analysis.

Institutional review board statement

The study was approved by the Research Ethics Committee of Zayed University (Application No. ZU23_074_F).

Data availability statement

The datasets generated and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Declaration of competing interest

We, Mohammad Amin Kuhail, Nazik Alturki*, Justin Thomas, Amal K. Alkhalif declare that we have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this publication.

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Appendix

Table 9
Counseling scenarios extracted from CBT books.

No.	Scenario	Reference
H1	Patient: Oh, I don't know. Everything is such a mess. I'm doing terribly at school. I'm way behind. I feel so tired and down all the time. I feel sometimes like I should just give up. Counselor: Have you had any thoughts of harming yourself? Patient: No, not really. I just wish all my problems would somehow go away. Counselor: It sounds like you're feeling overwhelmed. Patient: Yes, I don't know what to do. Counselor: Okay, it sounds like you have two major problems right now. One is that you're not doing well at school. The other is that you feel really tired and down. Are there any others? Patient: I know I'm alone in my room too much. I should be spending more time with my friends. Counselor: Okay, let's turn these problems into goals. How would you like to be different? Or how would you like your life to be different as a result of treatment? Patient: I'd like to be happier. Counselor: And if you were happier and not feeling depressed, what would you be doing? Patient: I guess I would get involved in some activities at school like I did last year. I'd have some fun and not feel so depressed all the time.	(Beck, 2011), Pages 68, 69
H2	Patient: I often feel angry at my partner. Counselor: Ah, let's explore your anger, which is a feeling that will really upset you and can easily wreck your relationship with your partner. When do you usually experience that feeling, real anger, at your partner? Patient: Oh, quite often, especially when he contradicts me in front of others. Counselor: You mean, in front of your other employees? Patient: Yes, in front of potential customers. I really hate that! Counselor: And do you feel angry at him for frustrating you like that? Patient: Yes, very angry at him for acting so badly and frustrating me so much.	(Ellis, 1995), Page 20
H3	Patient: I like school. Like, I don't want to not be in school. Counselor: OK. what about school do you like? What keeps you here on the university campus? Patient: Well, I have really good friends here. Like, I feel like I've made really good friends. And like, I don't want to have a bad job. Like, I want to get a degree so I can have a good job. Yeah. I don't know. Like, I think the town is pretty and the campus is pretty. And I mean, I'm pretty happy here. Counselor: Mm-hmm. And what do you think kept you from going to class as regularly as you did when you were in high school? What kept you attending your college classes on a regular basis? Patient: Because like, with high school, I had to be there at 7:45 every morning. It was always the same. But like now it's like, Tuesday mornings I can like, sleep in, but like, Wednesday mornings and Monday mornings, I have to get up early. And I don't know. I think it's really hard to adjust to that, because you don't have the same schedule all the time. Counselor: OK. So you need a little more consistency. You feel like that might help you.	(Sommers-Flanagan & Sommers-Flanagan, 2018), Page 91
H4	Patient: So, one thing that I've been noticing lately is that I have a tendency to kind of hang back in groups, and I tend to observe Instead of participate. Counselor: OK. Patient: And, a lot of times, I get really nervous when I feel like I'm expected to speak up. And I want to speak up, but I kind of feel like I freeze. Counselor: OK. And so in particular then, in groups, the setting is big groups, little groups? Does it matter? Patient: It doesn't matter a whole lot, but generally, I guess, medium-size—like, you know, in classes that I have. They're usually like maybe 12 people or so. So relatively small, medium. I don't know what size that is. Counselor: OK. And so but in your head, you probably would say, I would like to speak up, but, for some reason, you feel inside some anxiety or nervousness about that. Patient: Definitely.	(Sommers-Flanagan & Sommers-Flanagan, 2018), Page 105

(*continued on next page*)

Table 9 (*continued*)

No.	Scenario	Reference
	Counselor: OK. Patient: And then sometimes when I try to force myself to speak up when I actually don't feel ready, I get like, I feel like my throat's closing, and I get a red face. Counselor: OK. All right. So that's sort of one possible issue for issue for us to talk about. Could you maybe give me a specific example of a particular group situation, as recent as you can, where you had this feeling of hanging back and maybe some fear of evaluation.	

Table 10
Therapy scenarios developed by interacting with Pi.

No.	Scenario	Comparable Scenario from CBT books
A1	Patient: Oh, I don't know. Everything is such a mess. I'm doing terribly at school. I'm way behind. I feel so tired and down all the time. I feel sometimes like I should just give up. Counselor: Woah, slow down. That's a lot to unpack, but let's take a step back for a second. You're clearly not feeling your best, and you're struggling academically. That's totally normal and happens to a lot of students at some point. It's definitely not a reason to give up though. Can we talk about some ways to turn things around? Patient: Okay. Counselor: Okay, great. I know this might sound like a lot of work, but I think the first thing to do is to break it down into smaller, more manageable steps. Let's start with your classes. Can you tell me about one class in particular that's causing you the most trouble? Patient: Yes, calculus is a very hard class. Counselor: Calculus is definitely a tough subject, so I'm not surprised you're having a hard time. Let's tackle that class first. How are you doing in the class in terms of homework and test scores? Patient: Terribly. I think I will fail the class. Counselor: That's a tough spot to be in, but I'm confident we can turn it around. Let's start with the basics. How much time do you spend studying for this class every week? Patient: Very little time now because every time I start, I feel overwhelmed. I'm so behind.	H1
A2	Patient: I often feel angry at my partner. Counselor: Oh, that's a tough situation. I'm sorry to hear that. Can you tell me a bit more about what makes you angry with your partner? Perhaps I can help you work through the issue, or at least provide a sympathetic ear. Patient: He contradicts me in front of others. Counselor: That's really disrespectful. It can make you feel like your partner is undermining you. Is this a new habit or has it been going on for a while? Patient: It is an old habit. Counselor: Old habits die hard, but that doesn't mean they're impossible to break. How have you tried to address this issue with him in the past? Patient: Yes, without success.	H2
A3	Patient: I like school. Like, I don't want to not be in school, but I have not been going to class regularly. Counselor: That's a complicated situation. It's great that you enjoy school, but I can understand that there are probably some reasons why you're not attending classes regularly. Can I ask what some of those reasons are? Patient: It's hard to adjust to different schedules, like having to wake up early on some days but not others. High school had a set schedule, but now it's constantly changing. Counselor: Yeah, I can imagine that having such an irregular schedule can bne really disorienting and confusing. It's hard to build up a routine when everything changes so oftenn. And getting up early can be a challenge, especially if you're not a morning person. So I can understand why this might be impacting your attendance at school. Can I ask inf you're in college or a trade school? Patient: In college. Counselor: It's great that you're in college and getting an education, and managing that irregular schedule is even more difficult when you throw in all the other responsibilities that come with being in college. Is this your first year in college?	H3
A4	Patient: So, one thing that I've been noticing lately is that I have a tendency to kind of hang back in groups, and I tend to observe Instead of participate. Counselor: That's an interesting observation. Maybe you could try pushing yourself to participate more, and see how it goes? Or maybe you're just a more introverted type, and there's nothing wrong with that. Patient: Well, but, a lot of times, I get really nervous when I feel like I'm expected to speak up. And I want to speak up, but I kind of feel like I freeze. Counselor: That's really common! It's called "social anxiety," and it's pretty normal to feel nervous or uncomfortable when you're in social situations. It's almost like your brain just goes blank and you don't know what to say, even if you have something to say. Does that sound about right? Patient: Yes, I think so. Counselor: Okay, that's really insightful. So, if you're feeling that way, it's not because you're weird or flawed in some way. It's a common experience that a lot of people deal with, so please know that you're not alone in this. Can I ask if there are certain situations or types of people that trigger this more for you, or is it pretty consistent across all social situations?	H4

1. To what extent do you agree with the following statements regarding the counselor's performance?

	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
The counselor seemed trustworthy.	○	○	○	○	○
The counselor seemed warm.	○	○	○	○	○
The counselor seemed empathetic.	○	○	○	○	○
The counselor seemed organized.	○	○	○	○	○
The counselor seemed competent.	○	○	○	○	○
The counselor seemed interested in the conversation.	○	○	○	○	○

2. Based on this short extract, if that was you in the session, how satisfied would you have been?

	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
Based on this short extract, If that was you in the session, how satisfied would you have been?	○	○	○	○	○

3. I feel this transcript is (Select one option and provide a reason for your choice):
   ○ The product of human-human interaction (Human)
   ○ The product of human-artificial intelligence interaction (AI)

4. Reason for your choice:

Fig. 2. The questionnaire completed by the participants.

Data availability

Data will be made available on request.

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